Database Deployment

Læringsmål

- At kunne deploye en mysql database
- At kunne integrere en deployed database i en deployed Spring Boot applikation

Indhold

- Overblik over CI/CD pipeline
- (Repetition)
  - environment variabler
  - Spring profiles
- DB deployment
- DB integration

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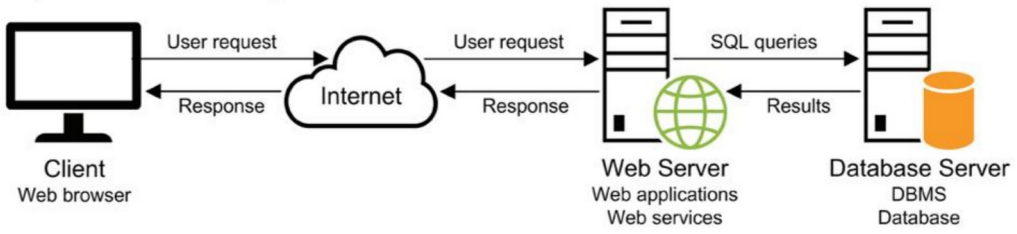
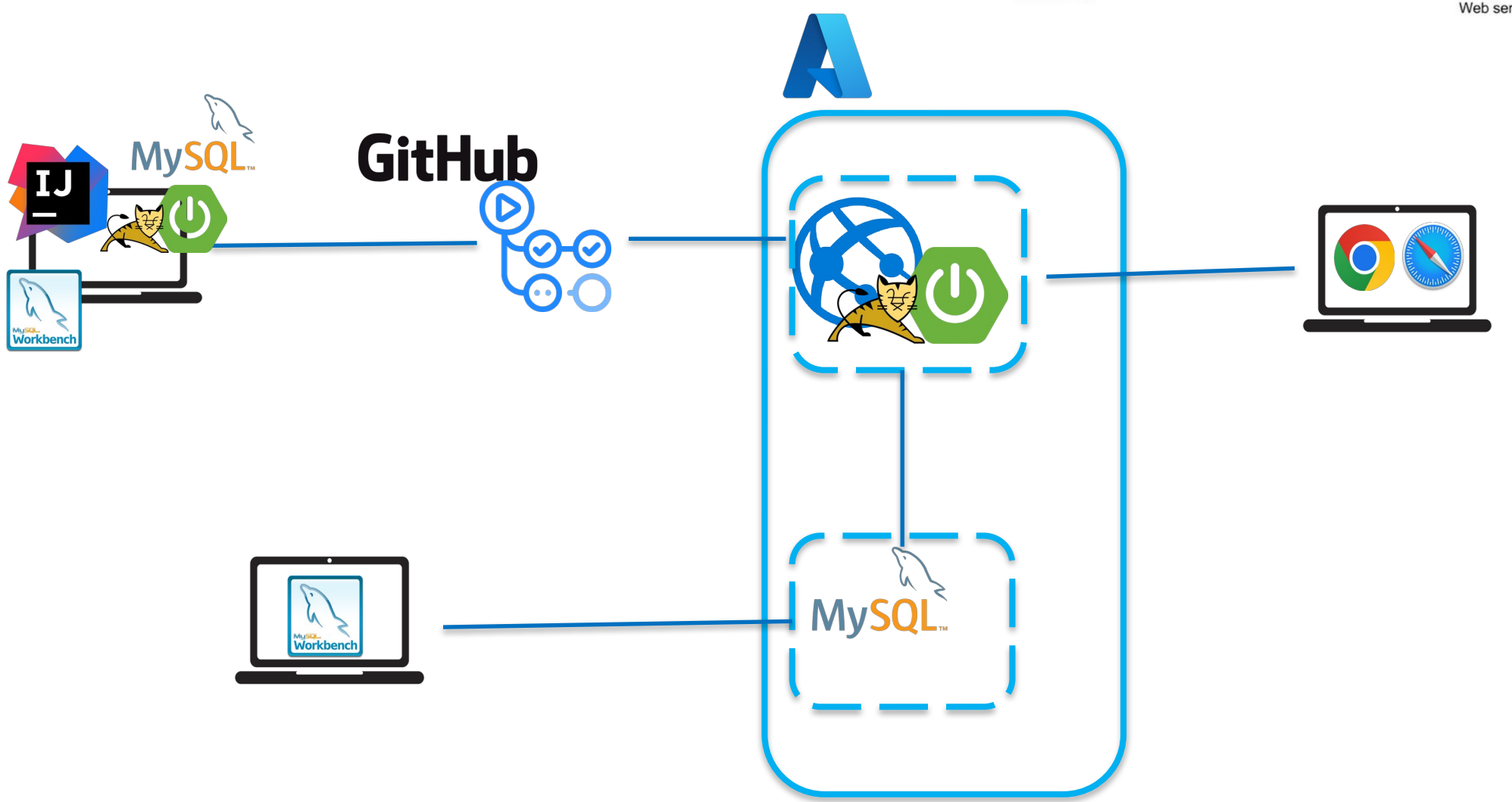
CI/CD Pipeline



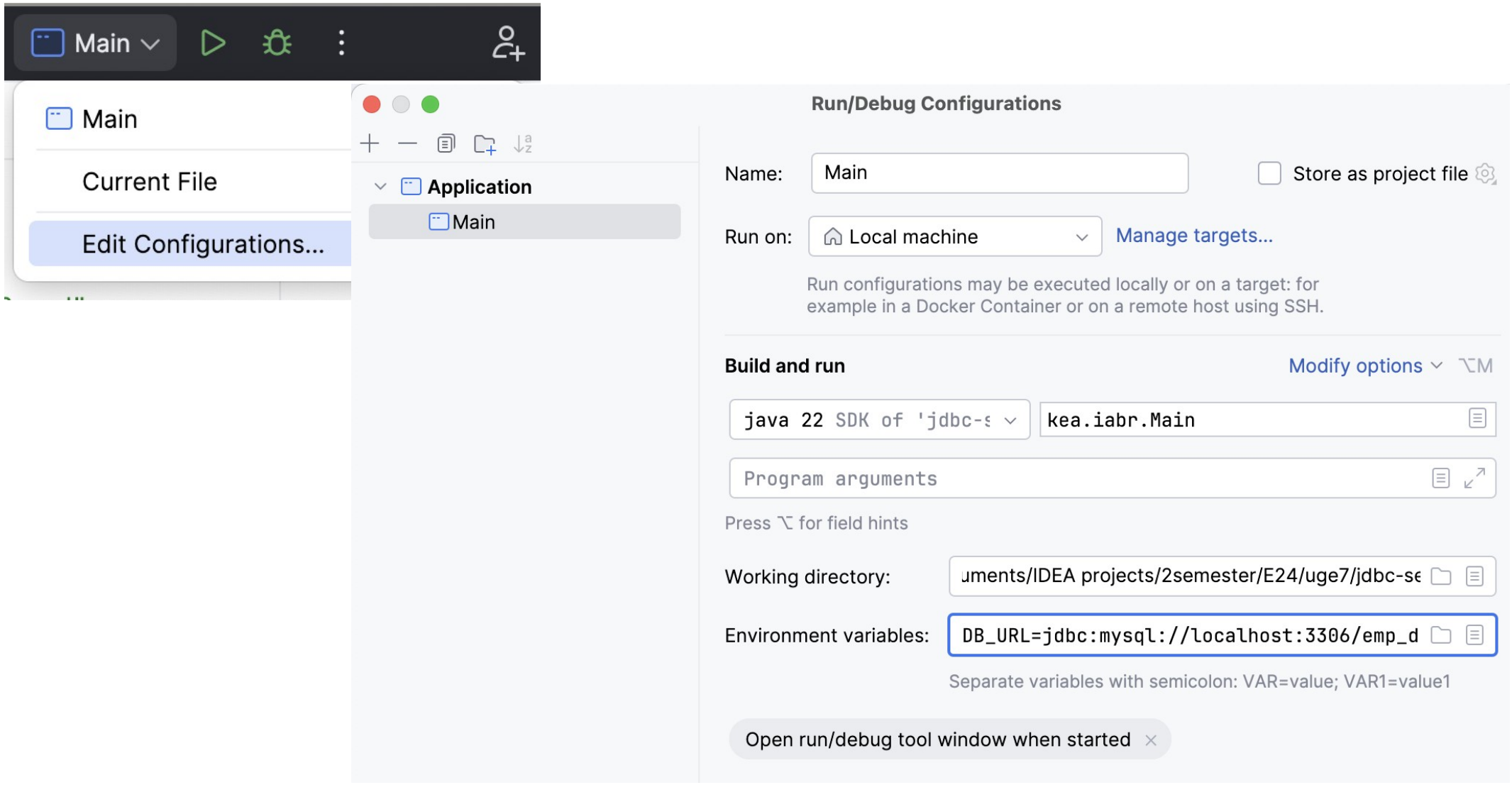


Environment Variables
- No Hardcoded Credentials

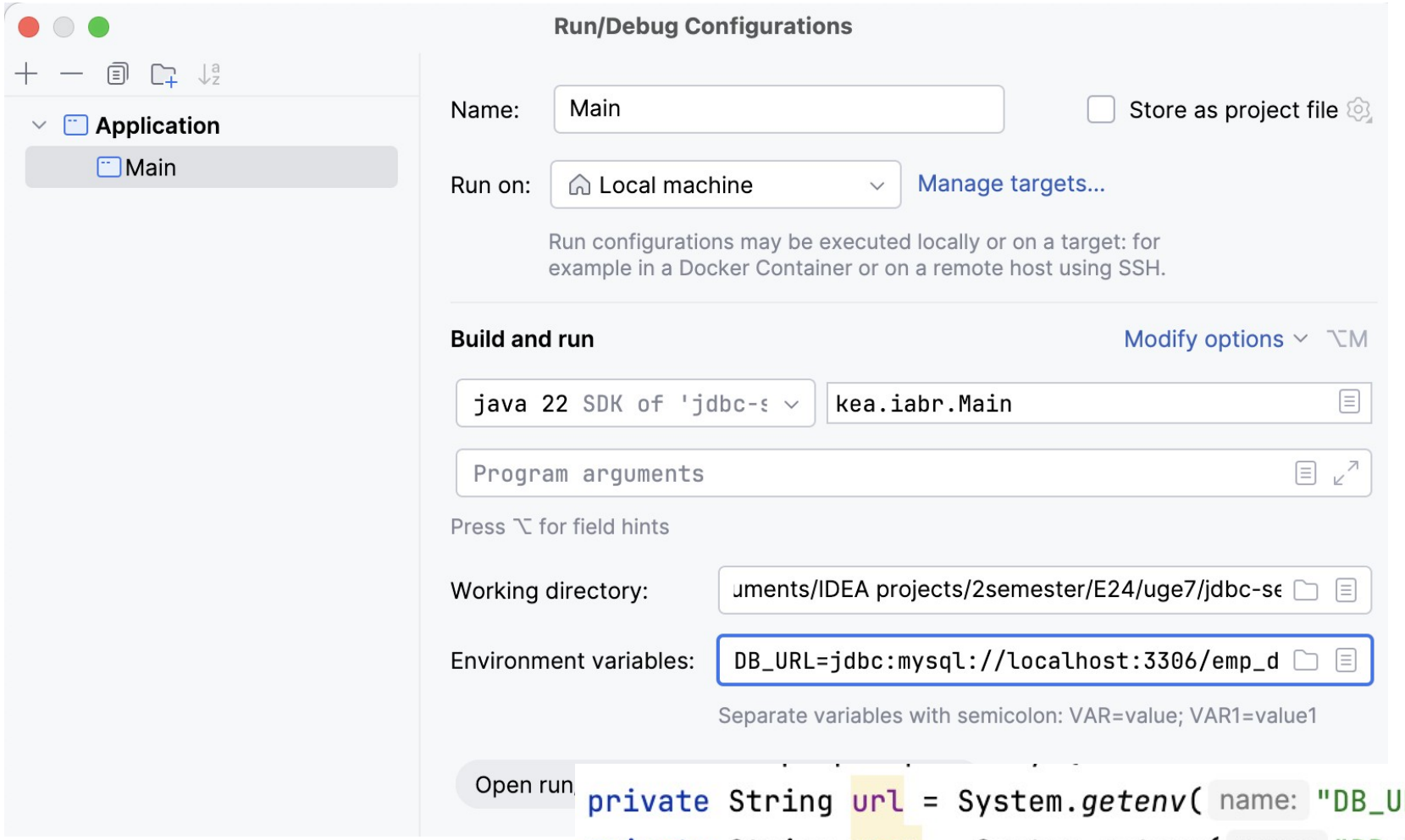
- Use Environment variables
  - key-value pairs stored by the operating system
  - used by running processes and applications to access system-wide or user-specific settings
  - provide a way to pass configuration data to programs without hardcoding the values into the application itself.

- used to store sensitive information (e.g. database credentials), system configuration (e.g. file paths or URLs), or runtime data that may change depending on the environment (development, testing, production)

Environment Variables in IntelliJ



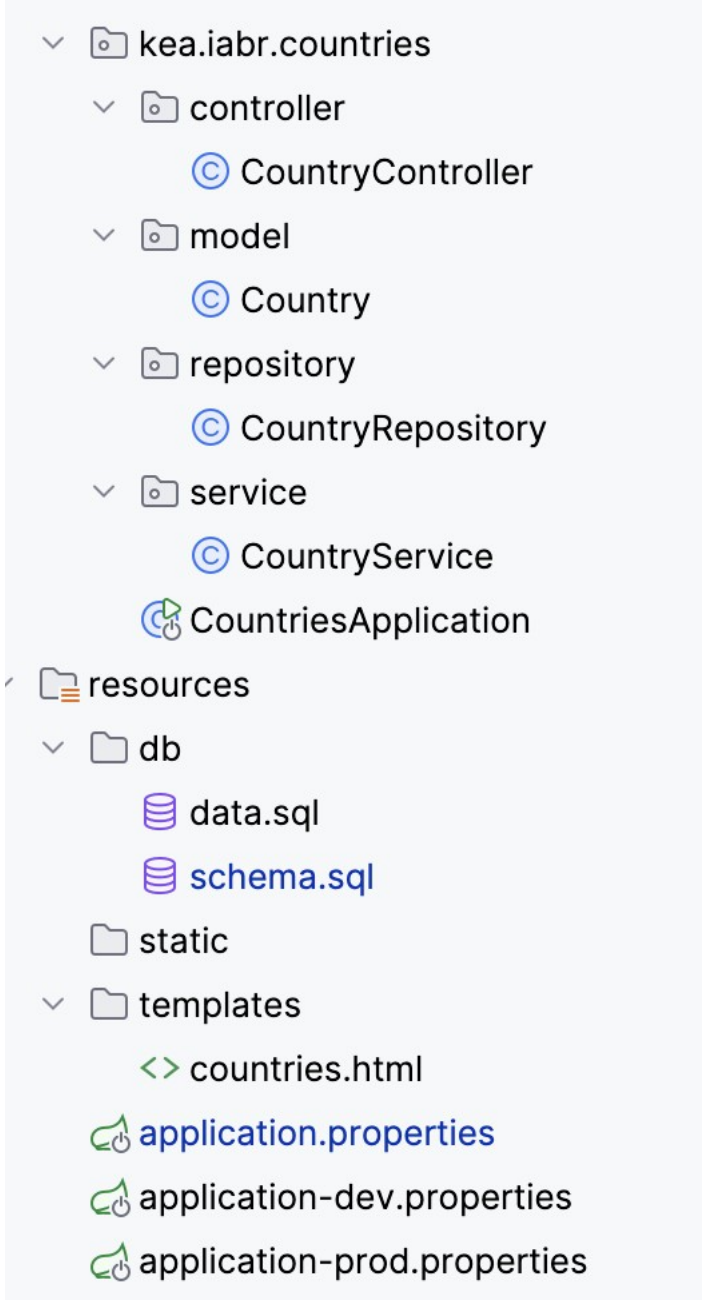
Environment Variables in Java




```
private String url = System.getenv( name: "DB_URL");
private String user = System.getenv( name: "DB_USER");
private String password = System.getenv( name: "DB_PASSWORD");
```

JDBC and Spring Boot

- application.properties file
- configuration file used to define settings
- used to provide environment variables (e.g. DB credentials) to the application



JDBC and Spring Boot -
environment variables and the application.properties file

- Define e.g. DB credentials as environment variables

Environment variables: `DEV_DATABASE_URL=jdbc:mysql://localhost:`

Separate variables with semicolon: VAR=value; VAR1=value1

- Define placeholder in the application.properties file

```
spring.datasource.url=${DEV_DATABASE_URL}
spring.datasource.username=${DEV_USERNAME}
spring.datasource.password=${DEV_PASSWORD}
```

- In the repository class, use @Value annotation to inject the value of into a variable

```
@Repository
public class CountryRepository {

    @Value("${spring.datasource.url}")
    private String dbUrl;
    @Value("${spring.datasource.username}")
```

`dbUrl` now has the value: `jdbc:mysql://localhost:`
(the rest of the url is not shown)

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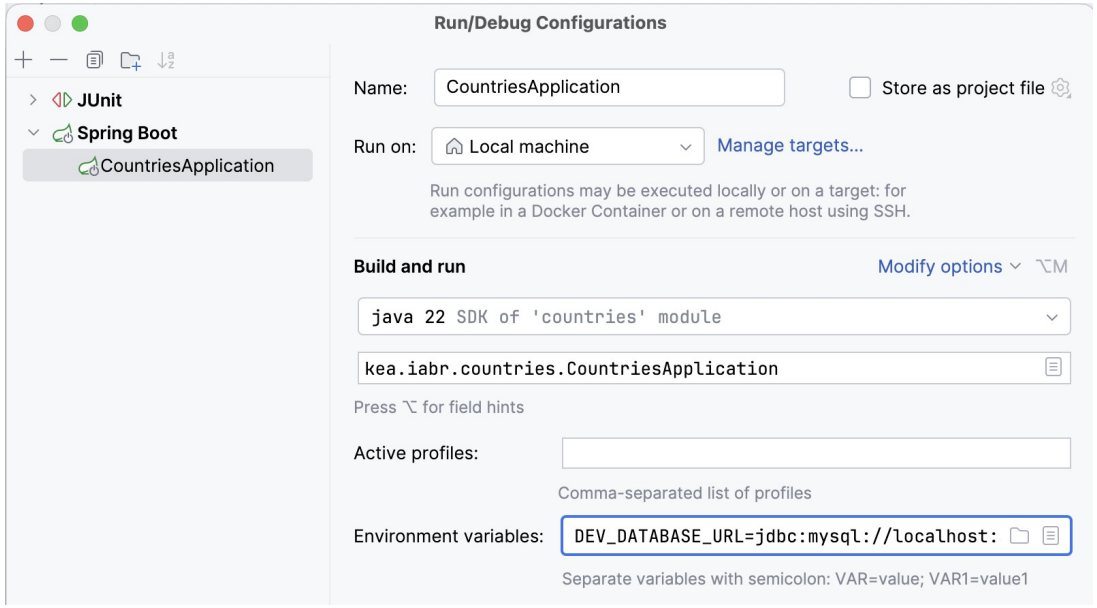
JDBC and Spring Boot -
environment variables and the application.properties file



application-dev.properties

```
1 spring.datasource.url=${DEV_DATABASE_URL}
2 spring.datasource.username=${DEV_USERNAME}
3 spring.datasource.password=${DEV_PASSWORD}
4
```

CountryRepository.java

```
10
11 @Repository
12 public class CountryRepository {
13     @Value("${spring.datasource.url}")
14     private String dbUrl;
15     @Value("${spring.datasource.username}")
16     private String username;
17     @Value("${spring.datasource.password}")
18     private String password;
```

Spring Profiles

- Use different configurations for different environments
  - e.g. use different databases for development, testing and production.

- Spring Profiles enables different configurations to be defined for different environments (dev, test, prod) and activate them based on the environment the application is running in.

- Environment-specific configurations can be used without modifying code by specifying the active profile.

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Spring Profiles

application.properties

```
spring.application.name=countries
spring.profiles.active=dev
```

application-dev.properties

```
spring.datasource.url=${DEV_DATABASE_URL}
spring.datasource.username=${DEV_USERNAME}
spring.datasource.password=${DEV_PASSWORD}
```

application-prod.properties

```
spring.datasource.url=${PROD_DATABASE_URL}
spring.datasource.username=${PROD_USERNAME}
spring.datasource.password=${PROD_PASSWORD}
```

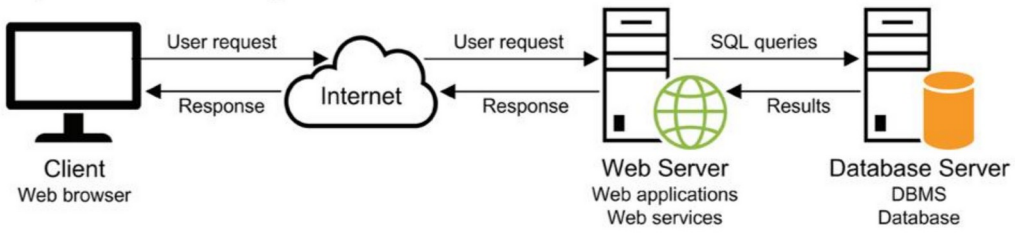
- kea.iabr.countries
  - controller
    - CountryController
  - model
    - Country
  - repository
    - CountryRepository
  - service
    - CountryService
  - CountriesApplication
- resources
  - db
    - data.sql
    - schema.sql
  - static
  - templates
    - countries.html
  - application.properties
  - application-dev.properties
  - application-prod.properties

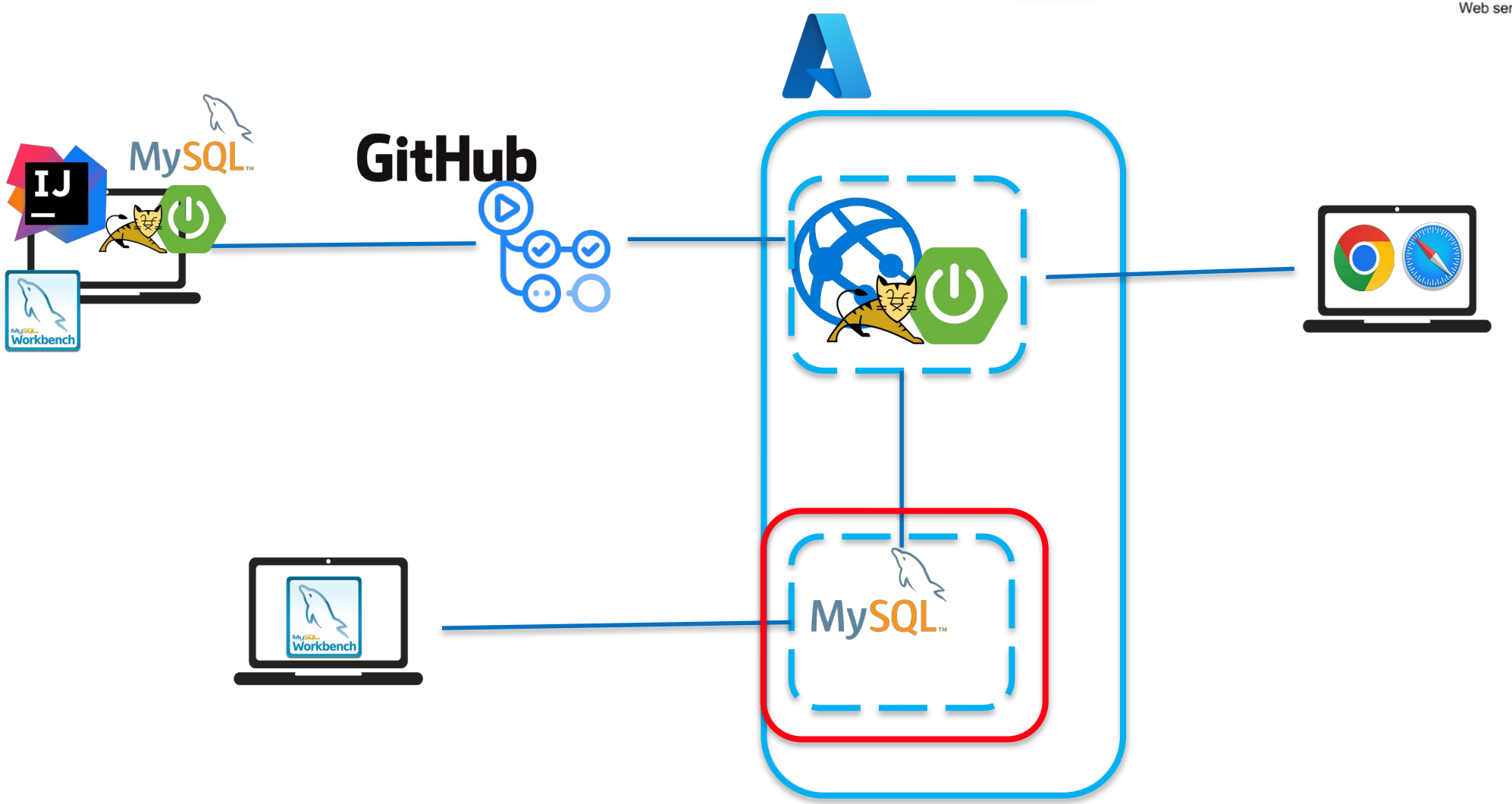
```java
@Repository
public class CountryRepository {

    @Value("${spring.datasource.url}")
    private String dbUrl;
    @Value("${spring.datasource.username}")
    private String username;
    @Value("${spring.datasource.password}")
    private String password;

    public List<Country> findAll() {
        List<Country> countries = new ArrayList<>();
        String sql = "SELECT * FROM countries";

        try (Connection connection = DriverManager.getConnection(dbUrl, username, password);
             Statement statement = connection.createStatement();
             ResultSet resultSet = statement.executeQuery(sql)) {

            while (resultSet.next()) {
                Country country = new Country(resultSet.getLong( columnLabel: "id"), resultSet.getString( columnLabel: "name"));
                countries.add(country);
            }

        } catch (SQLException e) {
            e.printStackTrace();
        }

        return countries;
    }
}
```

countries app - localhost

- countries db (countriesschema.sql, countriesdata.sql)
- https://github.com/GitHubBruger/countries.git

DB Deployment





DB Deployment

- Vejledning på Fronter

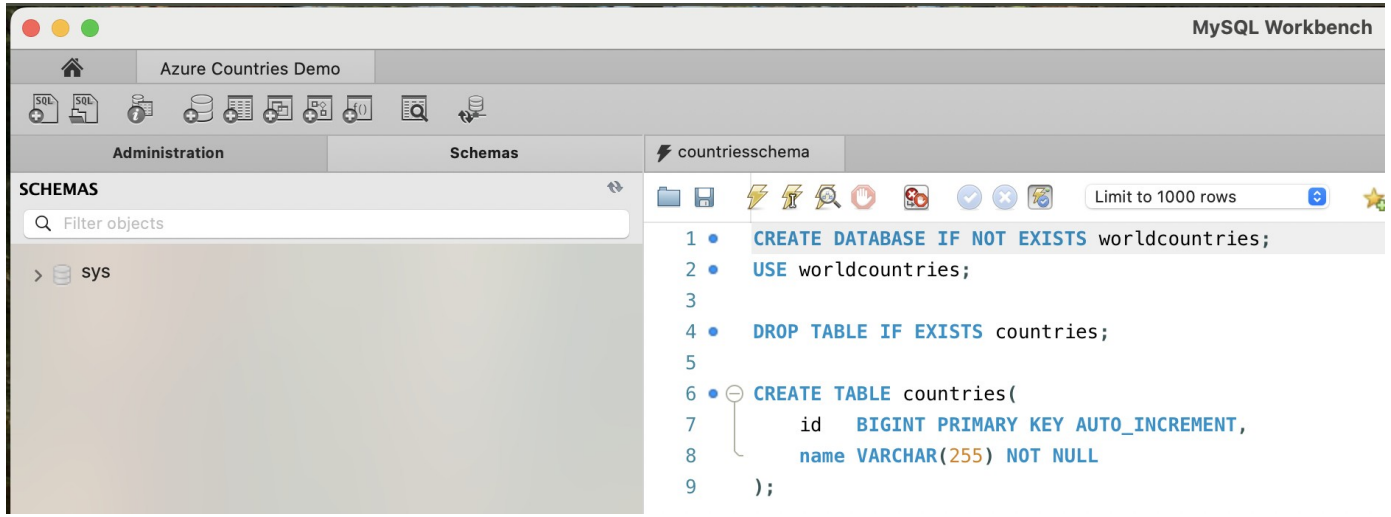
MySQL Database deployment på Azure

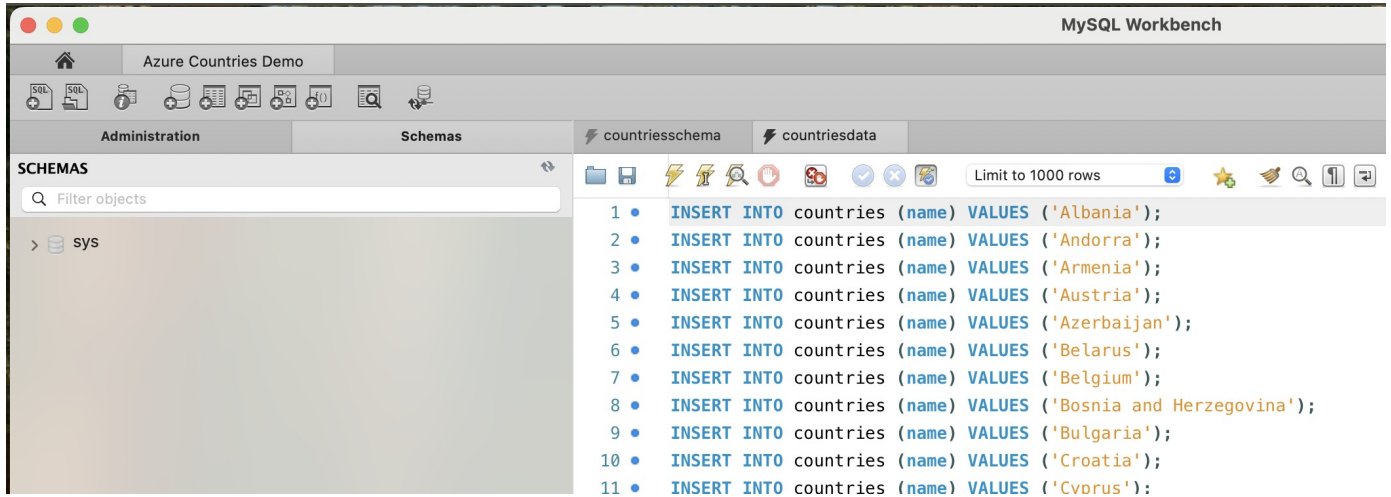
Denne guide beskriver hvordan man opretter en MySQL Database Server på Azure og efterfølgende opretter en MySQL database ved brug af f.eks. MySQL Workbench.
Det kræver at man har oprettet en konto på Azure og er logget ind på Azures portal (https://portal.azure.com/#home).

1. Vælg "Free services"

DB Deployment

MySql Workbench – deployed database





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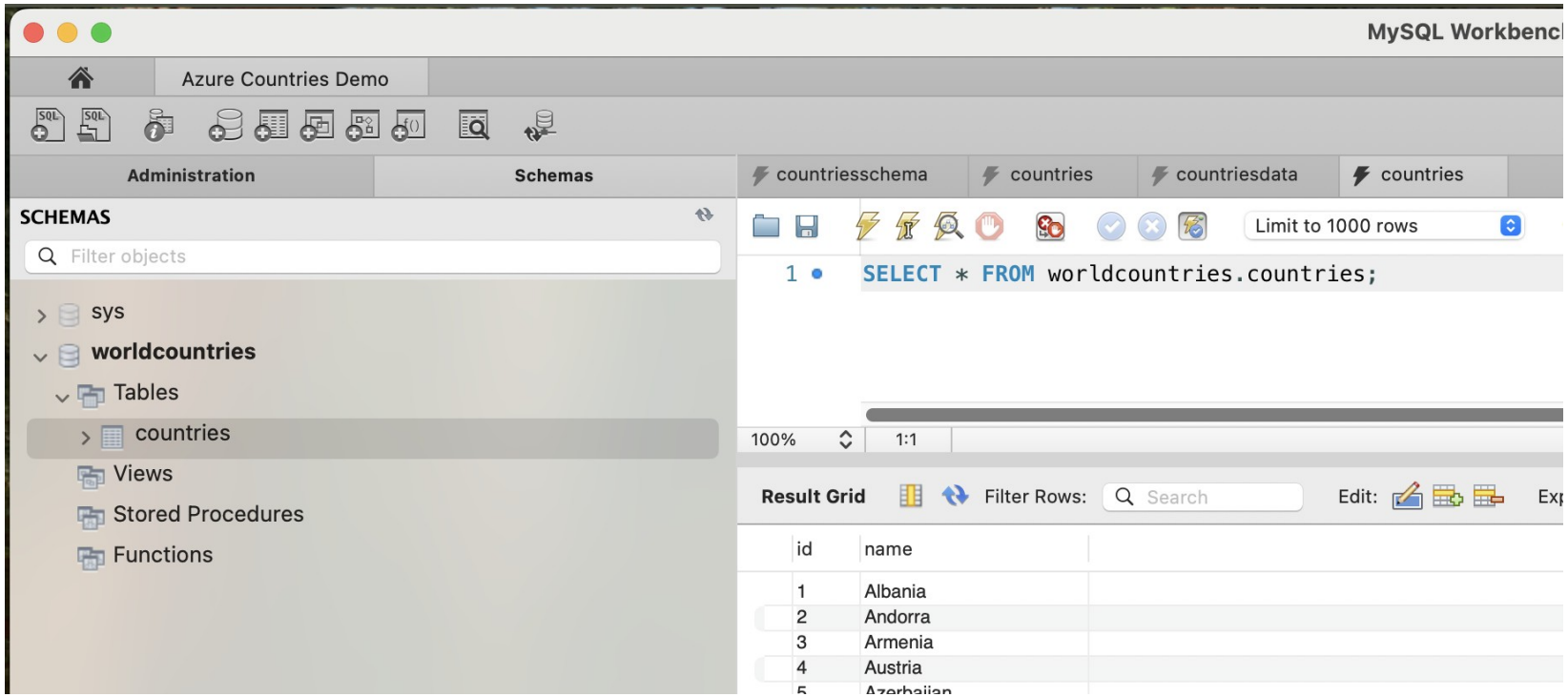
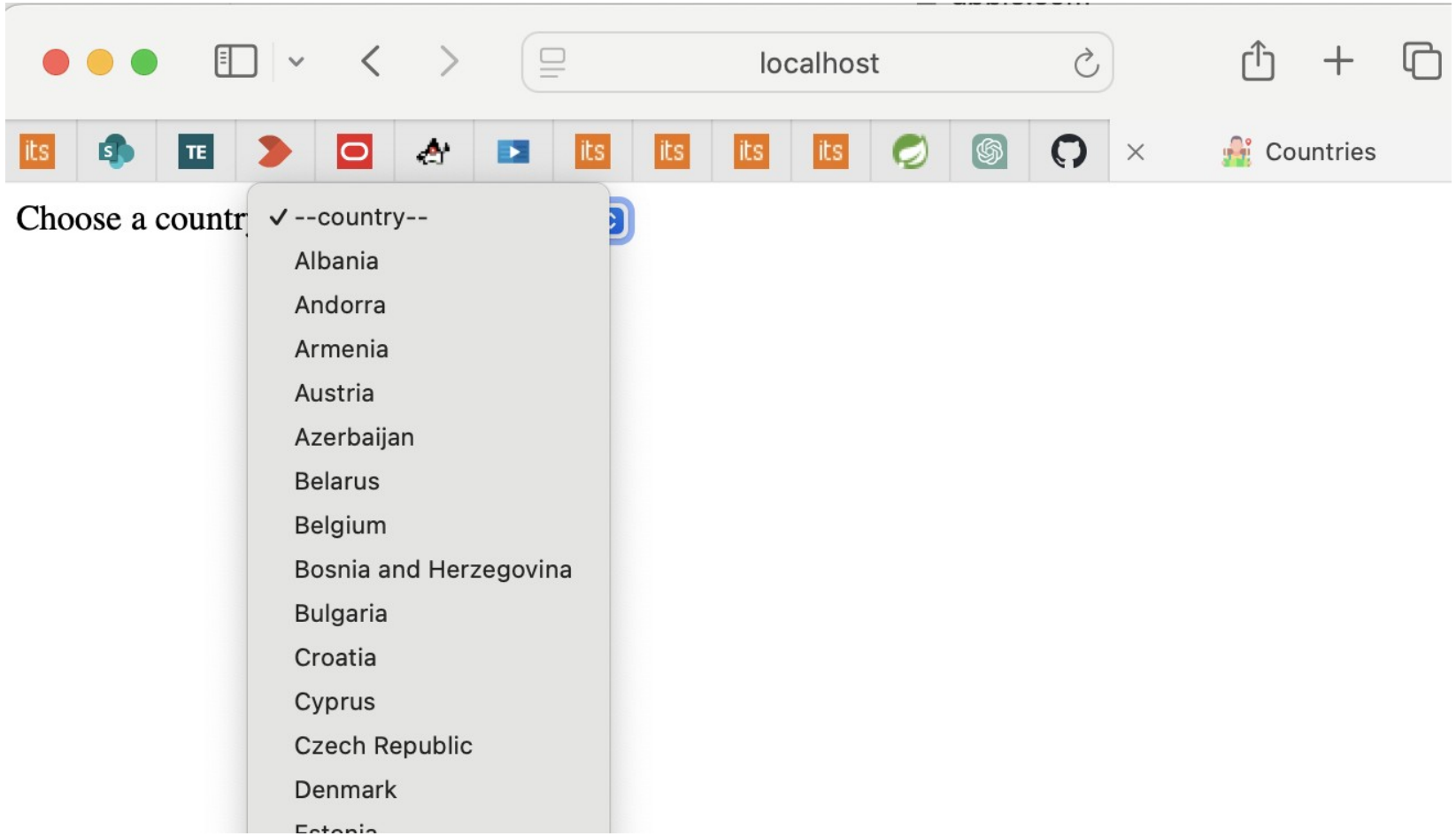
DB Deployment

MySql Workbench – deployed database



Localhost Spring Boot app med deployed DB



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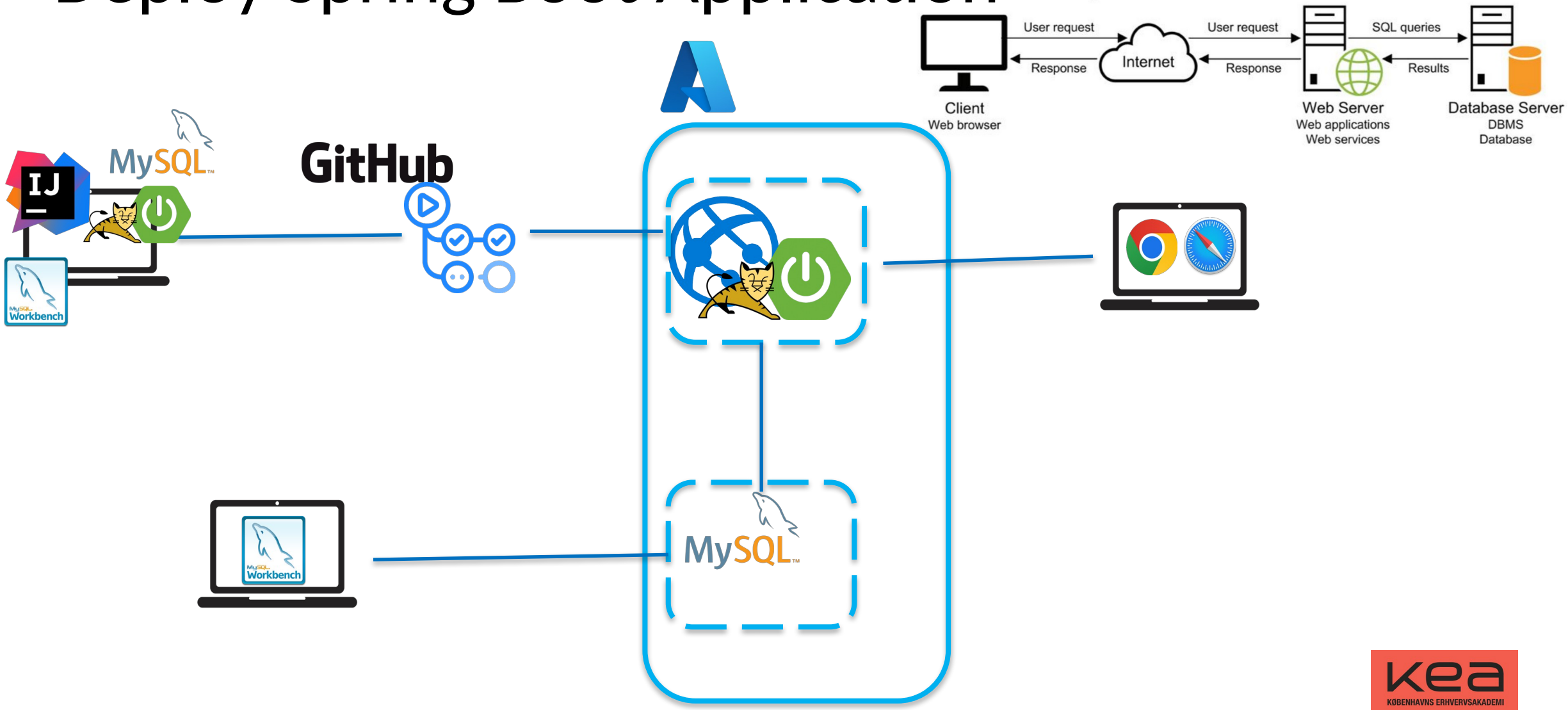
Deploy Spring Boot Application



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Spring Boot og JDBC Deployed

- Environment variables?
  - no longer localhost – need to define in prod/deployed environment (and GitHub)

application.properties

```
spring.application.name=countries
spring.profiles.active=prod
```

CountryRepository.java

```
@Repository
public class CountryRepository {
    @Value("${spring.datasource.url}")
    private String dbUrl;
    @Value("${spring.datasource.username}")
    private String username;
    @Value("${spring.datasource.password}")
    private String password;
```

Environment variables for the deployed app

1. Define environment variables on GitHub as GitHub secrets
2. Edit build job in workflow file to include GitHub secrets environment variables
3. Add environment variables to the web app on Azure Portal

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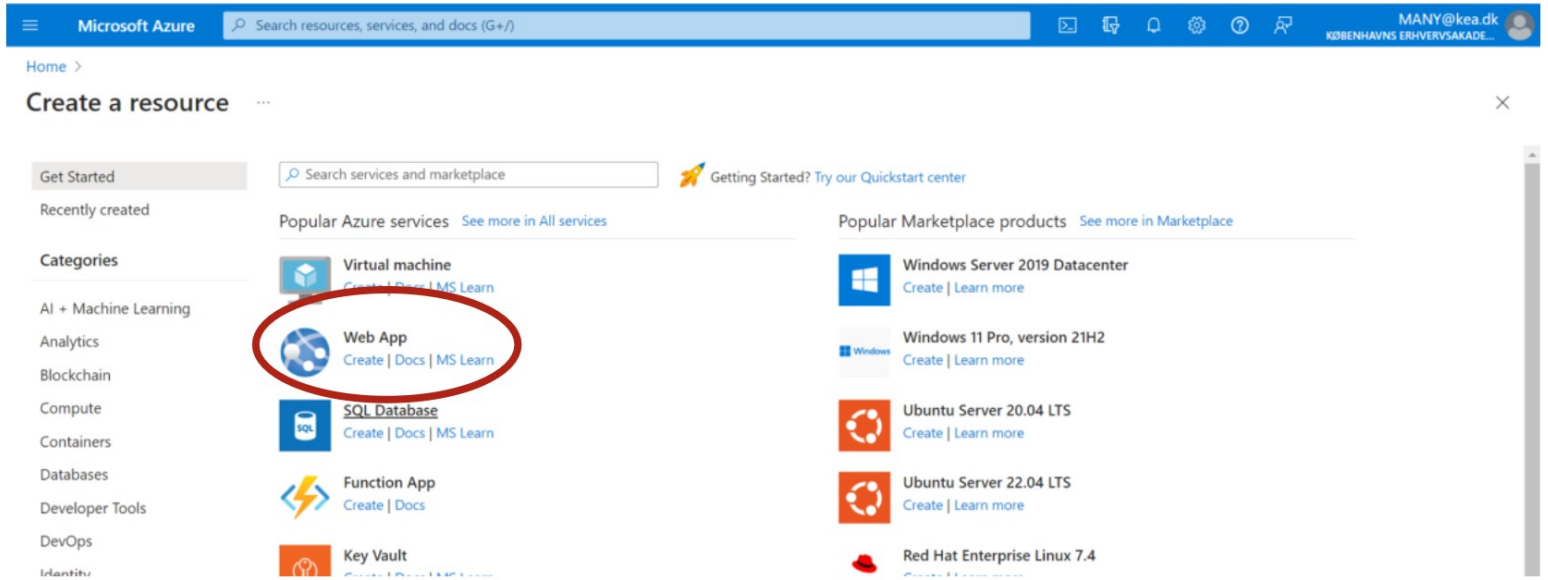
Deploying the app (repetition)

- Vejledning på Fronter (Deploy til Azure.pdf)



Udfyld felter.
Bemærk Instance name skal være globalt tilgængeligt.
Vigtigt: Vælg “Free F1” under Pricing plan

Deploying the web app

Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

Home > Create a resource >

Create Web App

Subscription * — Azure-sub-kea-ianb

Resource Group * — rg-countriesdemo

Create new

Instance Details

Name — country-demo

-gngadvgbd8hndkcf.northeurope-01.azurewebsites.net

Unique default hostname (preview) on. More about this update

Publish * — Code / Container / Static Web App

Runtime stack * — Java 17

Java web server stack * — Java SE (Embedded Web Server)

Operating System * — Linux / Windows

Region * — North Europe

Not finding your App Service Plan? Try a different region or select your App Service Environment.

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app.
Learn more

Linux Plan (North Europe) * — ASP-rgcountries-bb0b (F1)

Create new

Pricing plan — **Free F1** (Shared infrastructure)

Environment variables for the deployed app

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Define environment variables on GitHub as repository secrets

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | **Settings**

General

Access
- Collaborators
- Moderation options

Code and automation
- Branches
- Tags
- Rules
- Actions
- Webhooks
- Environments
- Codespaces
- Pages

Security
- Code security
- Deploy keys
- **Secrets and variables**
  - Actions
  - Codespaces
  - Dependabot

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. Learn more about encrypted secrets. Variables are shown as plain text and are used for **non-sensitive** data. Learn more about variables.

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

Secrets | Variables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository secrets

New repository secret

Name	Last updated
AZUREAPPSERVICE_CLIENTID_619C0E08BDC04303BAA3F9B1B4DE5D63	last month
AZUREAPPSERVICE_CLIENTID_4564711BBF1F45DEB9FC2334DA97CC3D	last month
AZUREAPPSERVICE_CLIENTID_C312E2A685B742EABBEE480F9A1C80E8	last month

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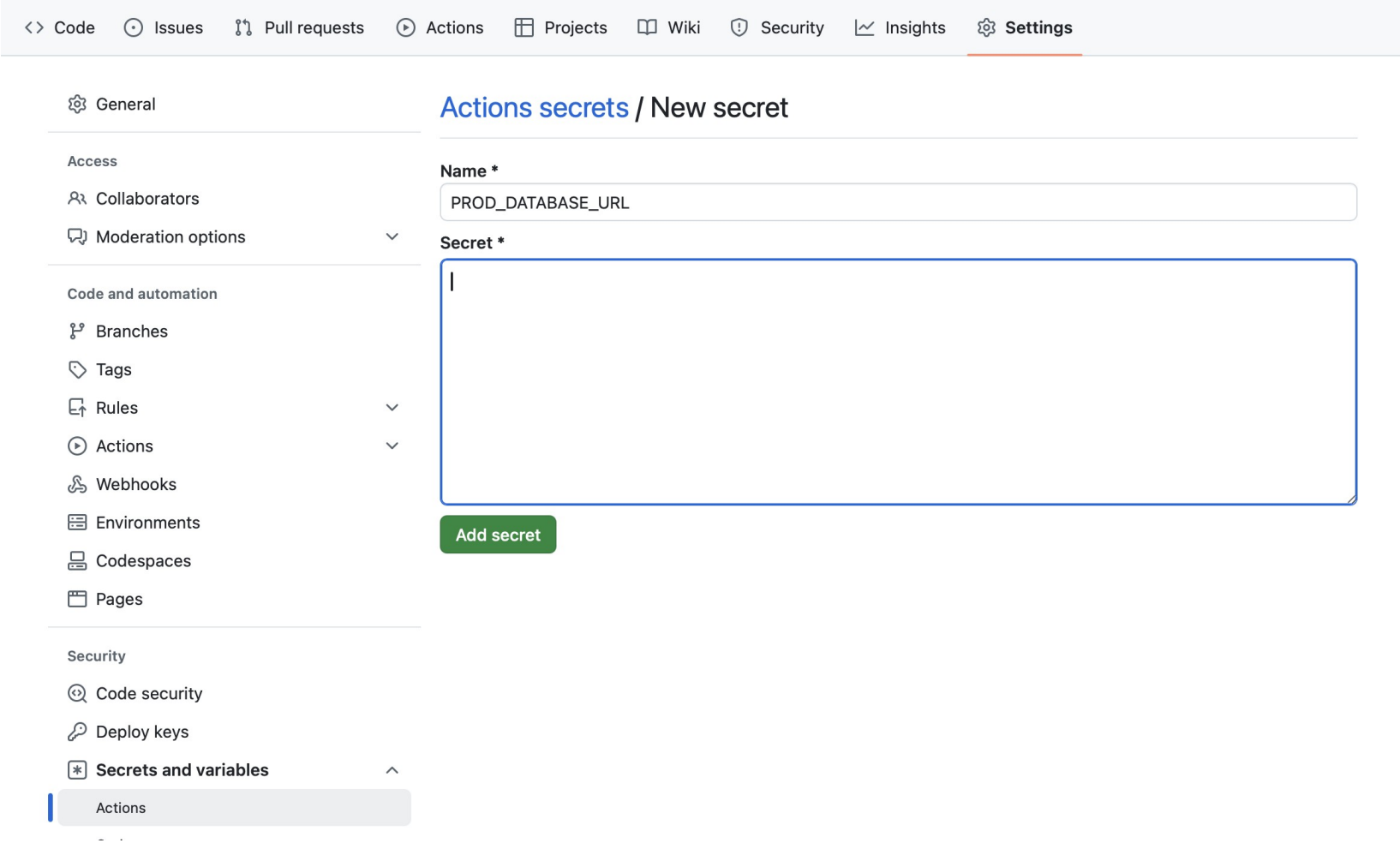
Define environment variables on GitHub as repository secrets



Environment variables for the deployed app

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Edit build job in workflow file to include GitHub Secrets environment variables

```yaml
jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v4

      - name: Set up Java version
        uses: actions/setup-java@v4
        with:
          java-version: '17'
          distribution: 'microsoft'

      - name: Build with Maven
        env:
          SPRING_DATASOURCE_URL: jdbc:mysql://${{secrets.PROD_DATABASE_URL}}
          SPRING_DATASOURCE_USERNAME: ${{secrets.PROD_DATABASE_USERNAME}}
          SPRING_DATASOURCE_PASSWORD: ${{secrets.PROD_DATABASE_PASSWORD}}

        run: mvn clean install
```

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Environment variables for the deployed app

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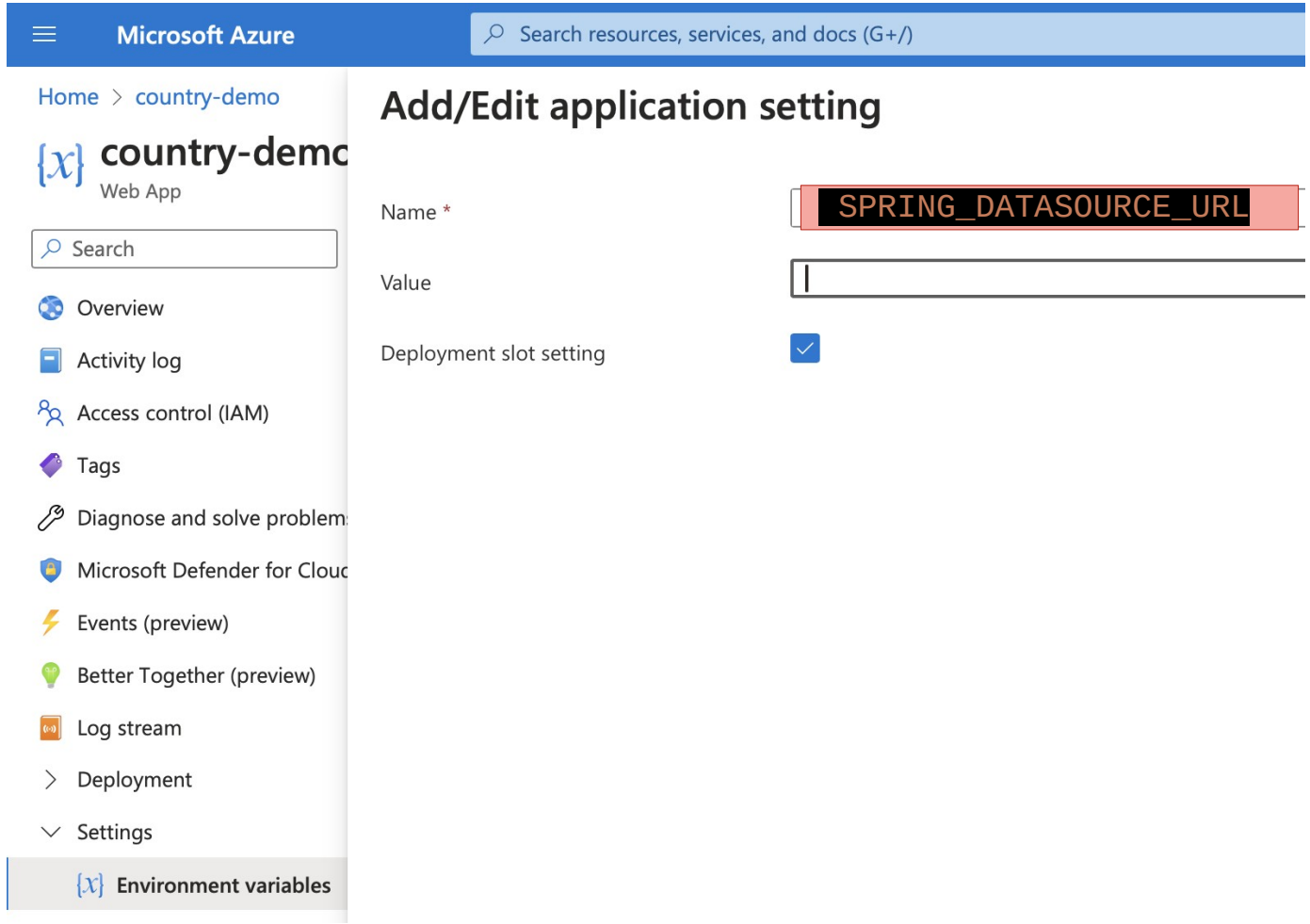
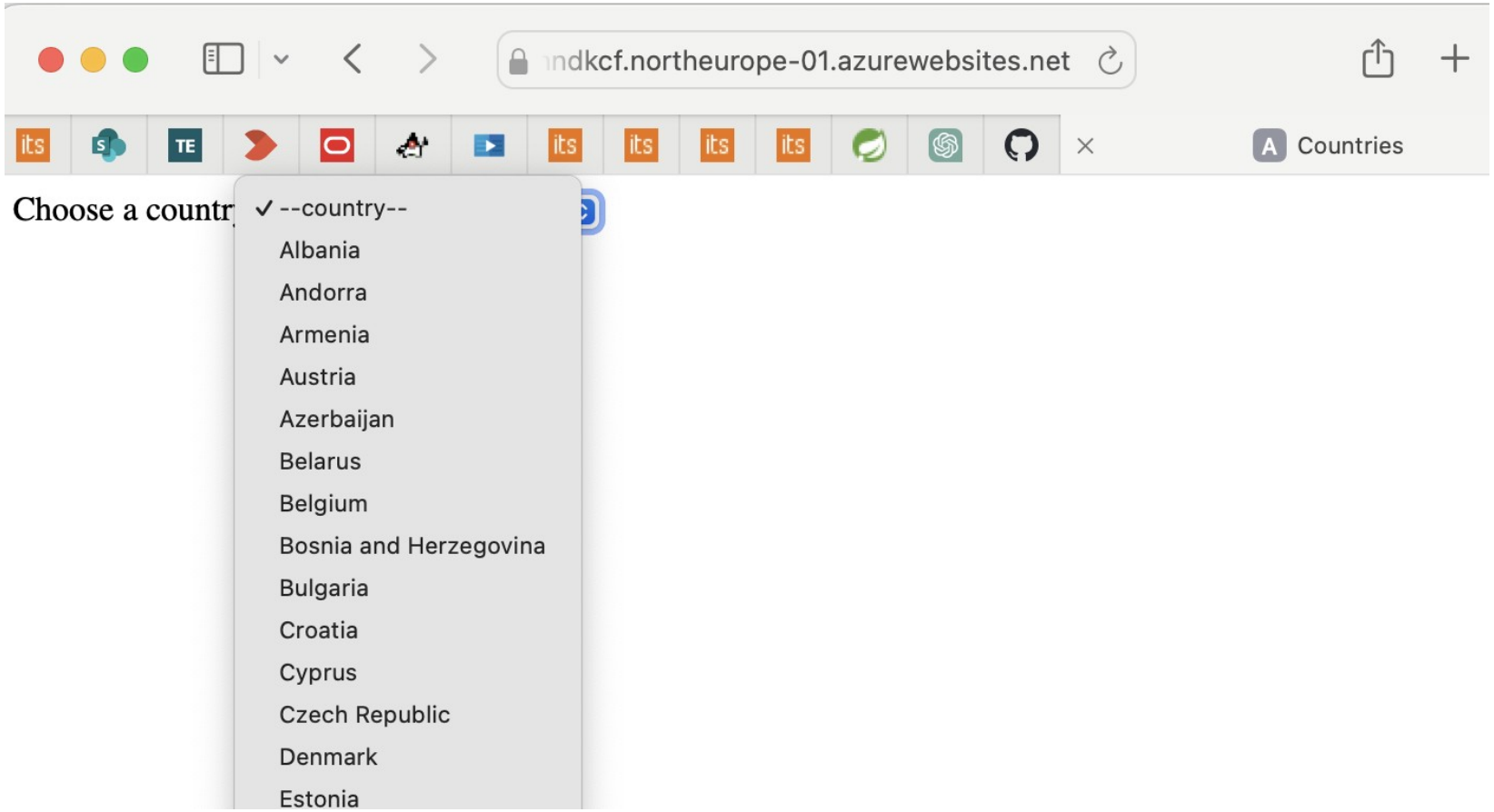
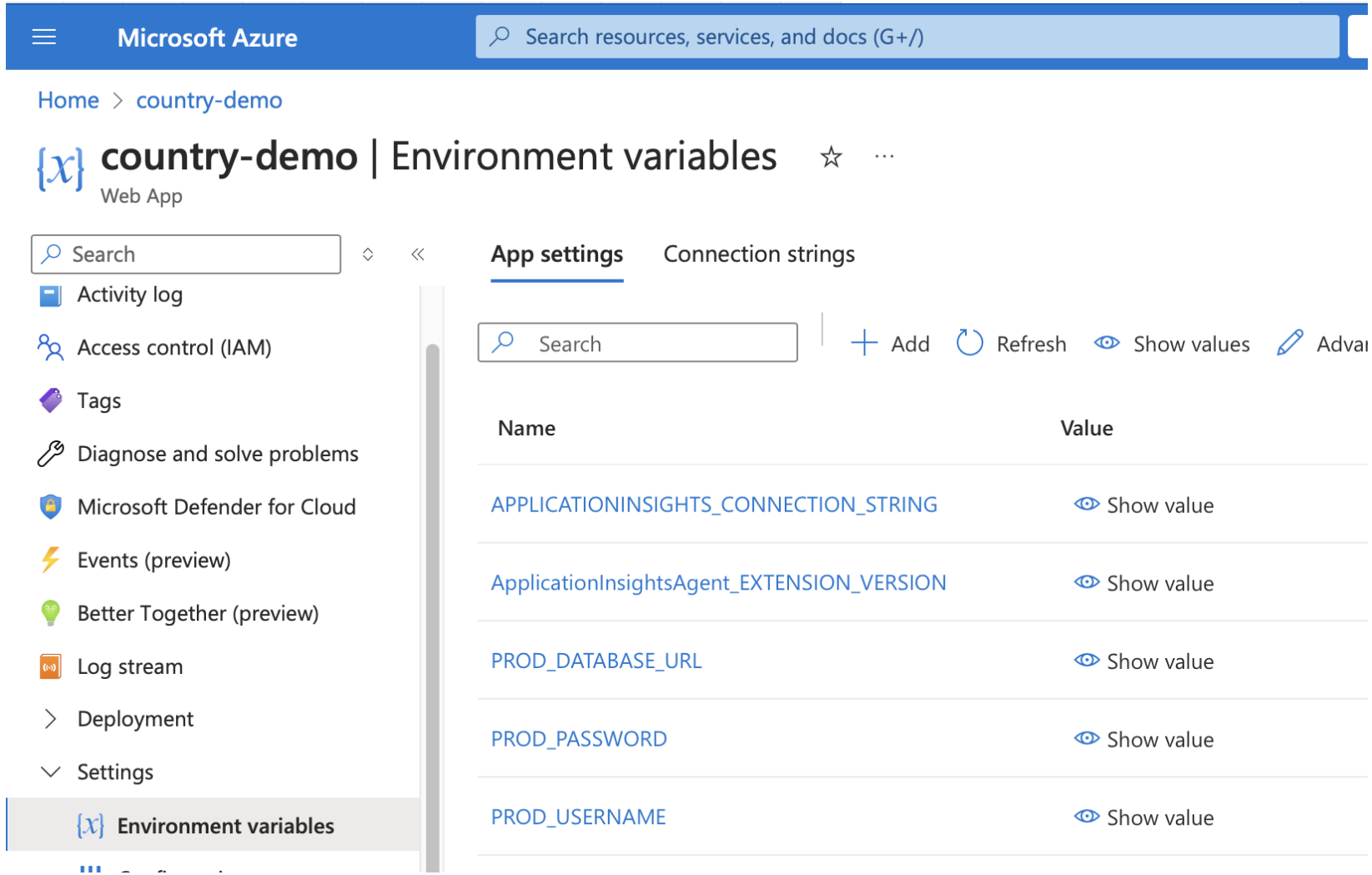
Add environment variables to the web app on Azure Portal



Deployed application and database



Add environment variables to the web app on Azure Portal



Environment variables for the deployed app

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Stop or Delete resources which are not being used

- NB. A stopped database server will start again automatically after 30 days